

Part A (BC): Graphing calculator required**Question 2****9 points****General Scoring Notes**

The model solution is presented using standard mathematical notation.

Answers (numeric or algebraic) need not be simplified. Answers given as a decimal approximation should be correct to three places after the decimal point. Within each individual free-response question, at most one point is not earned for inappropriate rounding.

A particle moving along a curve in the xy -plane is at position $(x(t), y(t))$ at time $t > 0$. The particle moves in such a way that $\frac{dx}{dt} = \sqrt{1+t^2}$ and $\frac{dy}{dt} = \ln(2+t^2)$. At time $t = 4$, the particle is at the point $(1, 5)$.

Model Solution	Scoring
(a) Find the slope of the line tangent to the path of the particle at time $t = 4$. $\left. \frac{dy}{dx} \right _{t=4} = \frac{y'(4)}{x'(4)} = \frac{\ln 18}{\sqrt{17}} = 0.701018$ The slope of the line tangent to the path of the particle at time $t = 4$ is 0.701.	Answer 1 point
Scoring notes: - To earn the point, the setup used to perform the calculation must be evident in the response. The following examples earn the point: $\frac{y'(4)}{x'(4)} = 0.701$, $\frac{\ln(2+4^2)}{\sqrt{1+4^2}}$, or $\frac{\ln 18}{\sqrt{17}}$. - Note: A response with an incorrect equation of the form “function = constant”, such as $\frac{y'(t)}{x'(t)} = \frac{\ln(18)}{\sqrt{17}}$, will not earn the point. However, such a response will be eligible for any points for similar errors in subsequent parts.	
	Total for part (a) 1 point

- (b) Find the speed of the particle at time $t = 4$, and find the acceleration vector of the particle at time $t = 4$.

$\sqrt{(x'(4))^2 + (y'(4))^2} = \sqrt{17 + (\ln 18)^2} = 5.035300$	Speed	1 point
The speed of the particle at time $t = 4$ is 5.035.		
$a(4) = \langle x''(4), y''(4) \rangle = \left\langle \frac{4}{\sqrt{17}}, \frac{4}{9} \right\rangle = \langle 0.970143, 0.444444 \rangle$	First component of acceleration	1 point
The acceleration vector of the particle at time $t = 4$ is $\langle 0.970, 0.444 \rangle$.	Second component of acceleration	1 point

Scoring notes:

- To earn any of these points, the setup used to perform the calculation must be evident in the response. For example, $\sqrt{(x'(4))^2 + (y'(4))^2} = 5.035$ or $\sqrt{17 + (\ln 18)^2}$ earns the first point, and $\langle x''(4), y''(4) \rangle = \left\langle \frac{4}{\sqrt{17}}, \frac{4}{9} \right\rangle$ earns both the second and third points.
- The second and third points can be earned independently.
- If the acceleration vector is not presented as an ordered pair, the x - and y -components must be labeled.
- If the components of the acceleration vector are reversed, the response does not earn either of the last 2 points.
- A response which correctly calculates expressions for both $x''(t) = \frac{t}{\sqrt{1+t^2}}$ and $y''(t) = \frac{2t}{2+t^2}$, but which fails to evaluate both of these expressions at $t = 4$, earns only 1 of the last 2 points.
- An unsupported acceleration vector earns only 1 of the last 2 points.

Total for part (b) 3 points

- (c) Find the y -coordinate of the particle's position at time $t = 6$.

$y(6) = y(4) + \int_4^6 \ln(2 + t^2) dt$	Integrand	1 point
	Uses $y(4)$	1 point
$= 5 + 6.570517 = 11.570517$	Answer	1 point
The y -coordinate of the particle's position at time $t = 6$ is 11.571 (or 11.570).		

Scoring notes:

- For the first point, an integrand of $\ln(2 + t^2)$ can appear in either an indefinite integral or an incorrect definite integral.
- A definite integral with incorrect limits is not eligible for the answer point.
- Similarly, an indefinite integral is not eligible for the answer point.
- For the second point, the value for $y(4)$ must be added to a definite integral.
- A response that reports the correct x -coordinate of the particle's position at time $t = 6$ as $x(6) = x(4) + \int_4^6 \sqrt{1 + t^2} dt = 11.200$ (or 11.201) instead of the y -coordinate, earns 2 out of the 3 points.
- A response that earns the first point but not the second can earn the third point with an answer of 6.571 (or 6.570).
- If the differential is missing:
 - $y(6) = \int_4^6 \ln(2 + t^2)$ earns the first point and is eligible for the third.
 - $y(6) = \int_4^6 \ln(2 + t^2) + y(4)$ does not earn the first point but is eligible for the second and third points in the presence of the correct answer.
 - $y(6) = y(4) + \int_4^6 \ln(2 + t^2)$ earns the first two points and is eligible for the third.

Total for part (c) 3 points

- (d) Find the total distance the particle travels along the curve from time $t = 4$ to time $t = 6$.

$\int_4^6 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$	Integrand	1 point
$= 12.136228$ The total distance the particle travels along the curve from time $t = 4$ to time $t = 6$ is 12.136.	Answer	1 point

Scoring notes:

- The first point is earned for presenting the correct integrand in a definite integral.
- To earn the second point, a response must have earned the first point and must present the value 12.136.
- An unsupported answer of 12.136 does not earn either point.

Total for part (d) 2 points

Total for question 2 9 points